

PHASE 1 – BRAINSTORMING & IDEATION

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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1. Introduction

In many organizations, different users should have access to different records based on their role, department, location, status, or other field values. Giving every user unrestricted access to all records can create security and privacy risks.

The proposed project aims to implement field-based access control using scripts and Access Control Lists (ACLs). The system will evaluate a specific field value in a record and determine whether the current user is allowed to access that record.

For example, employees belonging to the IT department may be allowed to view IT-related records, while employees from other departments may be restricted from accessing them.

2. Problem Statement

Traditional access-control mechanisms often provide access based primarily on user roles or groups. However, some applications require more detailed, record-level security.

For example:

- A user may have permission to access a particular table but should not see every record in that table.
- Employees may need access only to records belonging to their department.
- Managers may need access to records from their own department.
- Sensitive records may need to be restricted based on their status or classification.

Therefore, there is a need for a mechanism that dynamically evaluates record field values and grants or denies access accordingly.

3. Brainstorming

The following ideas were considered during brainstorming:

Idea 1 – Role-Based Access Control

Users are assigned roles such as:

- Administrator
- Manager
- Employee

Access is granted according to the assigned role.

Limitation: Role-based access alone may not provide sufficient record-level control.

Idea 2 – Department-Based Record Access

Users can access records only when the record's department matches the user's department.

Example:

1. User Department = IT
Record Department = IT
→ Access Allowed»
2. User Department = IT
Record Department = HR
→ Access Denied»

Idea 3 – Status-Based Access Control

Access can depend on a field such as:

- Open
- Closed
- Confidential
- Approved
- Pending

For example, confidential records could be restricted to authorized users

Idea 4 – Field-Based Dynamic ACL

A script evaluates one or more record fields and dynamically determines whether the current user has access.

This approach provides more flexible and granular security.

4. Selected Idea

The selected idea is:

Script-Controlled ACL – Restrict Record Access Based on Field Value

The system will use an ACL with a script-based condition to evaluate a field value in the requested record.

The ACL script will determine:

- Who is requesting access.
- What record is being accessed.
- What value is stored in the relevant field.
- Whether the user's attributes/permissions match the record.
- Whether access should be granted or denied.

5. Proposed Solution

The proposed solution introduces a dynamic security mechanism in which access to individual records is controlled using a scripted ACL.

Basic workflow

User → Requests Record → ACL Executes Script → Field Value Checked → Access Granted/Denied

The script will compare the appropriate user information with the selected record's field value.

Example

Suppose a record contains:

Record	Department	Assigned User
R001	IT	User A
R002	HR	User B
R003	Finance	User C

If User A belongs to the IT department, the ACL can allow User A to access R001 while restricting access to records belonging to other departments.

6. Objectives

The main objectives of the project are:

- To implement record-level access control.
- To restrict unauthorized users from viewing protected records.
- To use scripting for dynamic ACL evaluation.
- To control access based on field values.
- To improve application security.
- To demonstrate how ACLs can provide granular authorization.
- To test both authorized and unauthorized access scenarios.

7. Key Features

The proposed system will include:

7.1 Script-Based Authorization

A script will dynamically evaluate whether access should be granted.

7.2 Field-Based Restriction

Access will depend on a specific field value in the record.

7.3 Record-Level Security

Users will not necessarily receive access to every record in a table.

7.4 Dynamic Access Evaluation

The ACL will evaluate the access request rather than relying only on static permissions.

7.5 Unauthorized Access Prevention

Users who do not satisfy the defined conditions will be denied access.

8. Benefits

The project provides the following benefits:

- Improved security: Prevents unauthorized access to sensitive records.
- Granular control: Allows access to be controlled at the individual-record level.
- Flexibility: Conditions can be changed according to organizational requirements.
- Reduced data exposure: Users see only records they are authorized to access.
- Dynamic authorization: Access decisions can be made using record and user information.
- Better compliance: Helps organizations enforce internal data-access policies.
- Scalability: The approach can be extended to multiple tables and business scenarios.

9. Target Users

The proposed solution can be useful for:

- System Administrators

- Application Developers
- Managers
- Employees
- Security Teams
- Organizations managing sensitive business records

10. Possible Use Cases

Use Case 1 – Department-Based Access

Employees can access only records belonging to their department.

Use Case 2 – User-Based Access

A user can access records assigned specifically to them.

Use Case 3 – Confidential Records

Records marked as confidential can be restricted to authorized users.

Use Case 4 – Location-Based Access

Users can access records associated with their permitted geographical location.

Use Case 5 – Status-Based Access

Access can be controlled according to the status of a record.

11. Initial Technology Idea

The project can be implemented using an environment that supports Access Control Lists (ACLs) and server-side scripting, such as ServiceNow.

Potential technologies/components include:

- ServiceNow platform
- Access Control Lists (ACLs)
- Server-side JavaScript
- Tables and records
- User/role information
- Script-based authorization conditions

12. Expected Outcome

At the end of the project, the system should demonstrate that:

- Authorized users can access permitted records.
- Unauthorized users cannot access restricted records.
- The ACL script evaluates the required field value.
- Access decisions are made dynamically.
- The complete process can be demonstrated through a working project.

13. Conclusion

The Script-Controlled ACL project addresses the need for fine-grained record-level security. Instead of allowing access based only on broad user roles, the system evaluates the values stored in records and dynamically determines whether the current user should be allowed to access them.

This approach can help organizations protect sensitive information while providing authorized users with the data they need to perform their work.

Phase 1 Outcome:

The project idea has been identified, the problem and proposed solution have been defined, possible approaches have been considered, and the key objectives, features, benefits, users, and use cases have been established.